

SAMIA TASMIM

Computer Science & Engineering Undergraduate

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PROFILE SUMMARY

Computer Science & Engineering undergraduate with a hands-on, build-first approach to software — comfortable moving between systems-level C/C++, full-stack web development, and desktop application design. Has shipped projects spanning OOP simulation, game development, database-backed desktop apps, and full-stack web products with AI-assisted workflows. A merit scholarship recipient throughout school with an active interest in robotics, technical documentation and team work.

EDUCATION

International University of Science and Technology — B.Sc. in Computer Science & Engineering (Undergraduate)

Admitted to Islamic University of Technology (IUT) with a partial OIC Scholarship, awarded competitively to host-country students

Bangladesh Agricultural University High School & College

Secondary and Higher Secondary education

SCHOLARSHIPS & AWARDS

- Awarded PSC, JSC, SSC, and HSC scholarships throughout primary and secondary education, based on merit
- Partial OIC Scholarship — awarded competitively to host-country students upon admission to IUT
- Honorable mention at junior category in national physics Olympiad

TECHNICAL SKILLS

Languages: C, C++, Java, JavaScript, TypeScript, SQL

Frameworks & Libraries: React, Spring Boot, JavaFX, raylib/raygui, Firebase, Tailwind CSS, Three.js, MediaPipe

Databases & Tooling: MySQL, Flyway, Maven, Vite, Git/GitHub, Eclipse, IntelliJ IDEA

Concepts: Object-Oriented Design, MVC & DAO patterns, RESTful APIs, Full-Stack Development, AI-assisted coding

Real-time & Auth: Socket.IO, Firebase Auth, Google Maps API

PROJECTS

Mochi (react-spring-app) — *full-stack study-companion web app with a reactive virtual pet (work on progress)*

Tools: React 19, TypeScript, Vite, Tailwind CSS v4, Spring Boot, MySQL, Flyway, Firebase Auth, MediaPipe, Rive

- Built a full-stack study companion where a virtual pet reacts to study sessions, focus time, and in-room interactions, with an XP/coin progression system, daily goals, and a leaderboard.
- Implemented webcam-based focus tracking with MediaPipe and an animated character state machine (Rive, with a hand-drawn SVG fallback renderer) driven by study state.
- Designed the Spring Boot + MySQL backend (Flyway migrations, Spring Security, Firebase-authenticated REST API) behind a typed, validated React/TypeScript frontend.

MovieScout — *JavaFX desktop movie browser backed by MySQL (work on progress)*

Tools: Java, JavaFX, MySQL, Eclipse, Scene Builder

- Built a Netflix-style desktop app to browse a 22-movie catalog with genre filtering, title search, cast detail pages, and a persistent Watch Later list.
- Applied an MVC + DAO architecture separating backend data access (MovieDAO, WatchLaterDAO) from FXML-driven frontend controllers.
- Implemented a live dark/light theme switch and a single-Stage view-swapping navigation shell.

Campus Survival Game (gamingg) — *2D multi-room adventure game with mini-games, built on raylib*

Tools: C, raylib, raygui, custom text-to-speech module

- Developed a room-based 2D adventure in C/raylib combining a maze-navigation level, an endless-runner sequence, a graph/shortest-path puzzle, and a boss-battle mini-game.
- Built custom drag-and-drop and quiz-based mini-games (timed multiple-choice questions, ingredient-matching) with a hand-rolled GUI layer on raygui.
- Integrated a text-to-speech module and a custom animated button/menu system for game navigation.

Social Platform Simulation (oopproject) — *console-based social-media platform demonstrating core OOP design*

Tools: C++, file-based persistence, Makefile

- Modeled a Facebook-style platform (Users, Posts, Comments, Likes, Friends, Pages, Notifications, Messenger, News Feed) as a set of cooperating manager classes.
- Used a generic templated GenericManager<T> base class for shared CRUD logic across managers, applying inheritance and polymorphism throughout.
- Persisted all application state to flat text files, with an authentication service handling signup/login and session management.

Retro Love Photobooth — *browser-based retro photobooth with a Java backend*

Tools: Java (HTTP server), HTML, CSS, JavaScript (getUserMedia)

- Built a browser photobooth styled as a vintage Nikon camera, using getUserMedia for live webcam capture at up to 1080p with automatic resolution fallback.
- Implemented a 3-2-1 countdown with flash animation, in-camera photo review, retake/download options, and a lightweight Java backend to persist captures to disk.

Personal Portfolio (samia-portfolio) — *personal portfolio site with WebGL-driven visual effects*

Tools: React 19, TypeScript, Vite, Three.js, OGL, postprocessing

- Designed and built a personal portfolio site with custom animated components — a WebGL galaxy background, a pixel-dissolve section reveals, and a project carousel.
- Used Three.js/OGL with a postprocessing pipeline for real-time shader effects, driven entirely from custom React components.

EXTRACURRICULAR ACTIVITIES

IUT Robotics Society

- General Member, Publications Department
- Executive Member, PR & Documentation Subteam — Project Altair

INTERESTS

Web Development, machine learning, building side projects end-to-end and AI-assisted rapid prototyping.